

A literature-implied answer to the signed Fourier RKBS conjecture

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Status. *Literature-implied answer: full existential conjecture, with a stronger Wiener-class family.* This is an agent-identified application of a known feature-map theorem, not a claim of a new literature solution.

1 Source question

Xu and Ye’s *Generalized Mercer Kernels and Reproducing Kernel Banach Spaces* (arXiv:1412.8663) says in Remark 4.6, printed page 69 (PDF page 75):

“So, we conjecture that a non-positive definite function could be used to construct a RKBS by the Fourier transform.”

The decisive supporting result is Lin–Zhang–Zhang, arXiv:1901.01002, Theorem 2.3 (PDF pages 5–6). In abbreviated form, it says that Banach spaces W_1, W_2 with a continuous bilinear pairing and feature maps Φ_j whose spans separate the opposite space produce a pair of RKBSs, with reproducing kernel

$$K(x, y) = \langle \Phi_1(x), \Phi_2(y) \rangle_{W_1 \times W_2}.$$

2 Direct identification

We use the Fourier convention

$$\check{h}(t) = \int_{\mathbb{R}^d} h(\xi) e^{2\pi i t \cdot \xi} d\xi.$$

Proposition 1. *Let $h \in L^1(\mathbb{R}^d)$ be nonzero, real, and even, and set $\phi = \check{h}$. For every conjugate pair $1 \leq p, q \leq \infty$, the translation-invariant kernel*

$$K(x, y) = \phi(x - y)$$

is the reproducing kernel of a pair of (complex) RKBSs isometric to $L^p(E)$ and $L^q(E)$, where $E = \{\xi : h(\xi) \neq 0\}$. It also has a real form. In particular, sign-changing h give non-positive-definite examples.

Proof. On E , define

$$a = |h|^{1/q}, \quad b = \operatorname{sgn}(h)|h|^{1/p},$$

with the usual endpoint convention $r^0 = 1$ on E . Take $W_1 = L^q(E)$, $W_2 = L^p(E)$ and the bilinear pairing $\langle u, v \rangle = \int_E uv$. The feature maps

$$\Phi_1(x)(\xi) = a(\xi) e^{2\pi i x \cdot \xi}, \quad \Phi_2(y)(\xi) = b(\xi) e^{-2\pi i y \cdot \xi}$$

belong to W_1, W_2 , respectively, and Hölder's inequality gives a continuous pairing. Since $ab = h$,

$$\langle \Phi_1(x), \Phi_2(y) \rangle = \int_E h(\xi) e^{2\pi i(x-y)\cdot \xi} d\xi = \phi(x-y).$$

It remains only to check the separating-density hypothesis of Theorem 2.3. If $v \in L^p(E)$ annihilates every $\Phi_1(x)$, then $av \in L^1(E)$ and its Fourier transform vanishes everywhere. Fourier uniqueness gives $av = 0$ almost everywhere, hence $v = 0$ because $a > 0$ on E . The same argument applied to $ub \in L^1(E)$ proves the other density condition. Theorem 2.3 therefore supplies the two RKBSs and both reproducing identities.

For a real version, restrict $L^p(E)$ and $L^q(E)$ to the closed real subspaces of Hermitian functions $v(-\xi) = \overline{v(\xi)}$. Evenness of h makes both feature maps Hermitian, the pairing real, and the same Fourier-uniqueness argument nondegenerate. Thus the theorem applies over the real scalar field as well. $\square$

3 Explicit non-positive-definite kernel

For $d = 1$, take

$$\phi(t) = e^{-\pi t^2} - \frac{1}{2}e^{-\pi t^2/16}, \quad h(\xi) = e^{-\pi \xi^2} - 2e^{-16\pi \xi^2}.$$

Then $h \in L^1(\mathbb{R})$ is real, even, nonzero, and sign-changing: $h(0) = -1$, whereas $h(\xi) > 0$ for sufficiently large $|\xi|$. By the uniqueness part of Bochner's theorem, a continuous positive-definite ϕ cannot have this signed Fourier density. Hence ϕ is not positive definite, while the proposition makes $\phi(x-y)$ a two-sided reproducing kernel for every p -model above. This answers the source's existential conjecture.

4 Provenance and search bounds

The supporting authors do not mention Xu–Ye's conjecture; the relation is the direct identification above. The four run indexes and the parsed arXiv corpus were searched for the arXiv id and signed/non-PD Fourier RKBS terminology. Bounded arXiv web searches inspected the exact source, arXiv:1901.01002, and the nearby signed-measure Krein-space paper arXiv:2006.00247. No explicit all- p statement was found, but Theorem 2.3 already proves it after the displayed factorization. Literature provenance therefore controls the packet classification.

References

1. Y. Xu and Q. Ye, *Generalized Mercer Kernels and Reproducing Kernel Banach Spaces*, arXiv:1412.8663, Remark 4.6.
2. R. Lin, H. Zhang, and J. Zhang, *On Reproducing Kernel Banach Spaces: Generic Definitions and Unified Framework of Constructions*, arXiv:1901.01002, Theorem 2.3.
3. F. Liu, X. Huang, Y. Chen, and J. A. K. Suykens, *Fast Learning in Reproducing Kernel Krein Spaces via Signed Measures*, arXiv:2006.00247.